Reliability Engineering and System Safety 94 (2009) 1685–1693

Contents lists available at ScienceDirect

Reliability Engineering and System Safety

journal homepage: www.elsevier.com/locate/ress

Resiliency as a component importance measure in network reliability

John C. Whitson, Jose Emmanuel Ramirez-Marquez *

School of Systems & Enterprises, Stevens Institute of Technology, Castle Point on Hudson, Hoboken, NJ 07030, USA

ARTICLE INFO

Article history:
Received 21 January 2009
Received in revised form
16 April 2009
Accepted 7 May 2009
Available online 14 May 2009

Keywords:
Reliability
Two-terminal
Network
Resiliency
Component importance

ABSTRACT

This paper seeks to define the concept of *resiliency* as a component importance measure related to network reliability. Resiliency can be defined as a composite of: (1) the ability of a network to provide service despite external failures and (2) the time to restore service when in the presence of such failures. Although, Resiliency has been extensively studied in different research areas, this paper will study the specific aspects of quantifiable network resiliency when the network is experiencing potential catastrophic failures from external events and/or influences, and when it is not known a priori which specific components within the network will fail. A formal definition for Category I resiliency is proposed and a step-by-step approach based on Monte-Carlo simulation to calculate it is defined. To illustrate the approach, two-terminal networks with varying degrees of redundancy, have been considered. The results obtained for test networks show that this new quantifiable concept of resiliency provides insight into the performance and topology of the network. Future use for this work could include methods for safeguarding critical network components and optimizing the use of redundancy as a technique to improve network resiliency.

© 2009 Elsevier Ltd. All rights reserved.

1. Introduction

In the physical sciences, resiliency or elasticity refers to the ability of a substance or material to resume its natural shape after being distended by the application of forces. The elastic property of a material quantifies the degree of deformation that may occur before the material is unable to return to its original shape, or catastrophically, break [1]. Mathematically, this relationship is described in a stress–strain curve for a solid. Broadly, then, the physical resiliency concept has two components: (1) the material degree of deformation when a force is applied and, if deformation occurs, (2) the time to return to its original state.

The insight offered by this physical science description can be very useful to the systems engineer. Instead of a physical material, one can apply the resiliency concept to a system or a network by taking into account three key considerations: (1) the application of forces, (2) the degree of deformation, and (3) the capability to return to original shape. By taking into account these three key considerations, network resiliency should be related to the ability of a network to (1) provide a network function when under "the application of external forces" and if unable to do so, (2) restore the network function. Unfortunately from the perspective of this manuscript, and based on the extensive literature review undertaken for this research, resiliency, or the resilience of a network, is yet not clearly defined nor quantified in the context of network performance.

Although significant research has been performed on resiliency, when applied to specific network functions there is a problem regarding the concise definition of the concept. Currently, the concept of a resilient network can be mistaken or misunderstood for a network that is reliable or, survivable or, robust or, recoverable. For example, Westmark [2] in his extensive review for information system survivability reviews two definitions of survivability that are strongly similar to what resiliency defines, these are: (1) "the ability of a network to maintain or restore an acceptable level of performance during network failure conditions by applying various restoration techniques and the mitigation or prevention of service outages from potential network failures by applying preventative techniques" and (2) "that a system can be made robust to partially successful attack through general architecture features, through adaptability (flexible response to unanticipated changes) and flexibility (ability to adapt to a range of adverse events without having to anticipate the particular response in advance)." Moreover, and even most importantly, according to Wesmark [2] while most of the authors agree that "survivability" is highly important less than 1% of the articles he reviewed provide a computational approach.

The vagueness regarding the concept of resiliency illustrates that there is a need to clarify the concept and then, introduce computational techniques that quantify or support it. To address

Abbreviations: CIM, component importance measure; MANET, mobile ad-hoc network; RAW, reliability achievement worth; CCF, common cause failures

* Corresponding author. Tel.: +1 201 216 8003.
E-mail addresses: jmarquez@stevens.edu, Jose.Ramirez-Marquez@stevens.edu (J.E. Ramirez-Marquez).

0951-8320/$ - see front matter © 2009 Elsevier Ltd. All rights reserved.
doi:10.1016/j.ress.2009.05.001

Nomenclature

$G=(N, A)$	stochastic capacitated network where N represents the set of nodes and A the set of arcs
d	network flow requirement between source node s and sink node t
x_i	current state (capacity) of arc i
$\mathbf{b}_i$	$\mathbf{b}_i=(b_{i1}=0, b_{i2}, \ldots, b_{i\omega_i}=M_i)$ potential states for arc i
$\mathbf{p}_i$	$\mathbf{p}_i=(p_{i1}, p_{i2}, \ldots, p_{i\omega})$ Probability associated with each of the values taken by x_i (i.e. $p_{ij}=P(x_i=b_{ij})$)
$\mathbf{x}$	system state vector $\mathbf{x}=(x_1, x_2, \ldots, x_m)$ denotes the state of all the arcs of the network
$\varphi(\mathbf{x})$	network structure function representing capacity between s and t under system state vector $\mathbf{x}$
$R(\mathbf{x})$	network reliability $R(\mathbf{x})=P(\varphi(\mathbf{x})\geq d)$
$\Re_{(\alpha,\beta)}$	category I resiliency
α	number of component failures due to external causes
β	specific failure case containing α
$\Re_{\alpha}$	expected value of Category 1 resiliency for α
$M(m,n)$	uniform random number generator

the first issue, and based on the key considerations in the physical concept, network resiliency can be sorted into two main categories:

I. The sensitivity of network service to abnormal/external influences.
II. The sensitivity of network service restoration when in the presence of abnormal/external influences.

Rose [3] proposed a similar categorization for economic resilience. In his discussion, Category I (or static resiliency) is related to "*…the ability of an entity or system to maintain function (e.g., continue producing) when shocked…*" From a network perspective, the interest in Category I resiliency is then, on understanding the effect that external influences—herein understood as external causes of component failure—have on network service. Under this category, resiliency is tightly related to the network time to failure.

Similarly, Rose [3] describes Category II (or dynamic resiliency) as related to "*…the speed at which an entity or system recovers from a severe shock to achieve a desired state…*" From a network perspective, this category relates to the concept of time to restore services due to external causes of component failure. And, thus, resiliency is related to the network time to repair acceptable performance.

Thus, from the perspective of this manuscript resiliency is a composite of these two categories. That is, quantifying network resiliency is a two-step process that should first, describe the network's ability to tolerate external causes of component failure and second, based on this understanding, focus on quantifying the ability of the network to restore performance.

The main contribution of this paper is to provide an approach to quantify Category I resiliency in the context of networks. The specific networks under consideration are two-terminal networks that contain multiple nodes and links. Two-terminal networks allow to analyze systems that provide some kind of service between a specified set of nodes [4], usually called source and sink, where the source node can be thought of as the origin, and the sink node as the destination. For these networks, performance is quantified based on their ability to process such a service.

Specifically, the paper considers both binary and multi-state two-terminal networks as described in reliability research [4–6]. From a reliability perspective, two-terminal network analysis is based on quantifying the probability that the source and sink nodes can communicate (i.e. a binary case where the network and its components either work or fail [4]) or the probability that a required flow—generally defined as a network source–sink demand—can be satisfied from the source node to the sink node (i.e. a multi-state case that is a mixture of source–sink connectivity and capacity where the network and its components can have multiple "performance" states [6,7]).

At this point it is important to describe the main difference between the concepts of network reliability and Category I resiliency. Reliability quantifies the probability that the network performs its intended function, for a specific mission time, under normal and known operating conditions [8]. As defined in this paper, Category I resiliency quantifies the probability that the network performs its intended function, for a specific mission time, when in the presence of external causes of component failure–potential catastrophic failures due to attacks, disasters, etc. Thus, the main difference between these two concepts is the failure source. In reliability, network failure sources are internal and due to the wear, the tear or the intrinsic life of the network components. In Category I resiliency, the failure sources are external and internal; external, because components may cease to function due to man-made or natural events (events not considered under the normal and known operating conditions) and internal, because even under the external events the components adhere to their intrinsic life characteristics.

To quantitatively address the definition of Category I resiliency, this paper uses concepts from the area of reliability component importance measures (CIM). The concept of reliability importance is related to the sensitivity of the network function to changing conditions. In reliability, CIM quantify the sensitivity of network reliability to component failure [8,9] and have been widely used for identifying system weaknesses and to prioritize reliability improvement activities [10–15]—a task of high importance in large complex networks where component criticality is not immediately obvious to the designer. For example, systems engineering practice stresses the importance of examining component reliability early on in the engineering process for identifying critical components and for performance optimization via redundancy allocation and/or replication. Detailed reviews of CIM can be found in Ramirez-Marquez and Coit [10,11], Song and Der Kiureghian [16], Zio and Podofillini [12,13], and Rausand and Hoyland [9].

However, most CIM are based on single perturbation points (i.e. failure of one component) to describe the impact on network reliability and do not completely encompass Category I resiliency. Although, common cause failures (CCF) can be described as events that lead to simultaneous failure of multiple components due to a common cause, they are usually considered and estimated as part of the system reliability analysis [17,18]. That is, CCF when necessary should be included as a potential cause of failure within the operating environment. Thus, CCF are generally not implemented as CIM but as part of the reliability modeling. In Butler [19] the concept of cut set importance identifies component criticality based on the total number of distinct minimal cut sets of specific cardinality containing a specific system component. However, although obtained from a group analysis (i.e. cut set) this CIM provides insight about single component impact (i.e. the importance of the component based on its appearance on cut sets). Finally, it is important to note that recently Podofillini and

Zio [20] reference the problem of considering single perturbation points. To address this problem, the authors considered searching for the most important component combinations impacting system unavailability.

In accordance with Category I resiliency, this paper introduces a new CIM that quantifies, for a particular combination of external components failures, the probability that the network is able to function. In probabilistic terms, this CIM describes the distribution of two-terminal network reliability when the network experiences a number of simultaneous failures of size α. Moreover, by enumerating all the unique combinations of network components for any given α one can identify the component combinations that generate the worst network Category I resiliency. From a network level perspective, this new CIM allows recognizing the overall sensitivity of the network to external failures and completely encompasses the definition of Category I resiliency.

The remainder of this paper is organized as follows: Section 2 describes related work to the taxonomy of resiliency. Section 3 presents a discussion and mathematical procedure for the modeling of the computation of resiliency. Section 4 presents an illustrative example and the step-by-step simulation procedure to quantify it, followed by several binary and multi-state networks that are imitative of different applications and their simulation results. Section 5 presents our conclusions and future directions.

1.1. Assumptions

1. Component states are statistically independent.
2. The structure function $\varphi(\mathbf{x})$ is coherent. That is, improvement of component states cannot damage the system.
3. Component states and associated probabilities are known.
4. Arcs are all directed.

2. Literature review

The purpose of this literature review is to describe the state-of-the-art regarding the taxonomy proposed for network resiliency. Although notional considerations have significantly contributed to the interest and research in resiliency (Hollnagel et al. [21] provide a comprehensive discussion), the interest of this paper is on research that specifically provides approaches to quantify resiliency in networks. Unfortunately, due to the vagueness regarding the concept of resiliency, most of these techniques have been elusive to the review. For example, the literature review provided by Castet and Saleh [22] illustrates that in engineering-based applications the concept of resiliency and survivability are interchangeably used according to the specific context of a given study. Moreover, the authors illustrate that no definitions or models are unanimously adopted within the technical and scientific community.

In terms of quantification, Castet and Saleh [22] developed a "survivability" tool for space systems. Via stochastic Petri nets for a particular configuration of a space-based network they demonstrated the capability of the tool, using Monte-Carlo simulations to generate representative results of stochastic behavior of the spacecraft endogenous anomalies and failures. Their definition of survivability for the space-based network is similar to the Category I resiliency proposed in this paper.

As in [22], the research described in [23–27] uses term network survivability as a synonym for Category 1 network resiliency. In these studies, survivability is generally defined as the capability of the network to maintain service continuity in the presence of failures. Given that from their perspective the networks experience very few failures, the maximum number of failures considered is relatively low (in most studies a maximum of 2). So, for example She et al. [23] describe an approach for maximizing the probability of communication between two nodes assuming there is a failure probability of the network links. The problem is equivalent to the optimization of two-terminal reliability under a two independent path constraint. Banner and Orda [24] extend this problem by considering non-disjoint paths. Pullan [25] describes the efficiency of a flow network as a ratio of a network experiencing no failure to that of the network considering a single failed element. Zolfaghari and Kaudel [26] describe a framework for ensuring that networks maintain service despite failures. The framework is based on potential failure scenarios and various "survivability" metrics are proposed such as: the fraction of flow that remains after a failure scenario and, the fraction of flow that is restored after some given time. Bu et al. [27] evaluate the performance of Radio Access Networks under different failure scenarios (nodes and links) and propose resilient topologies that suffer minimum degradation in performance during failures, while requiring few additional links. Liew and Lu [28] describe survivability as the probability that a network can operate despite the event of a failure. The events are considered probabilistically and contain up to four component failures. Kalyoncu and Sankur [29] estimate the ability of a telecommunications network to maintain services despite failures. The failures are generated via the simulation of a damage model. Their results show that maintaining such services is highly dependent on the behavior of the damage model.

For electric power systems, Gautam et al. [30] implement a survivability index that considers line failures (i.e. contingencies) that discontinue the flow of power. The index is implemented to provide insight on how to combat contingencies. From their perspective, this is a challenging problem given that contingencies can happen even when there exists multiple layers of protective schemes in the power system. Al-Ammar and Fisher [31] define network resiliency as a ratio of network output to input. According to these authors, resiliency provides a ratio of the "performance" of the network when all nodes/links are in service to that with certain nodes/links taken out of service.

In summary, although there is significant amount of research done for quantifying the effects of component failures on network performance, the models developed quantify effect based on pre-specified damage models that cannot completely characterize Category 1 resiliency as defined in this paper. Moreover, under the quantification umbrella of this review, techniques for Category II resiliency were not found.

3. Resiliency modeling & computation

The rationale behind the definition of Category I resiliency (i.e. the probability that the network performs its intended function, for a specific mission time, when in the presence of external causes of component failure) is to establish a network performance profile describing the ability of the network to tolerate (or not) external causes of component failure. Moreover, as defined, Category I resiliency describes the distribution of network reliability when introducing external sources of component failure into the originally designed network. The insight gained by such profile is in identifying components combinations that significantly contribute to the ability or lack thereof of the network to perform its function and, based on this profile potentially, improving the overall Category I resiliency via redundancy or system reconfiguration.

3.1. Network description and resiliency modeling

Let $G = (N, A)$ represent a stochastic capacitated network with a required flow d between known source node s and sink node t. N represents the set of nodes and $A = \{i | 1 \le i \le m\}$ represents the set of arcs. The current state (capacity) of arc i, is defined by x_i which takes values from vector $\mathbf{b}_i$, represented by $\mathbf{b}_i = (b_{i1} = 0,\ b_{i2}, \ldots, b_{i\omega_i} = M_i)$ where $b_{ij} \in Z^+$, M_i equals the maximum capacity of arc i and ω_i equals its number of states. That is, vector $\mathbf{b}_i$ represents the range of states (capacities) associated with arc i. The vector $\mathbf{p}_i = (p_{i1},\ p_{i2}, \ldots, p_{i\omega})$ represents the probability associated with each of the values taken by x_i (i.e. $p_{ij} = \mathrm{P}(x_i = b_{ij})$). The system state vector $\mathbf{x} = (x_1,\ x_2, \ldots, x_m)$ denotes the state of all the arcs of the network. Function φ maps the system state vector into a system state. That is, $\varphi(\mathbf{x})$ is the network capacity between s and t under system state vector $\mathbf{x}$. The function $\varphi(\mathbf{x})$ can be analyzed for any possible realization of $\mathbf{x}$. For binary networks, $\varphi(\mathbf{x})$: $B^{|A|} \to B^+$ and thus, $R(\mathbf{x}) = P(\varphi(\mathbf{x}) = 1)$. For multi-state networks, $\varphi(\mathbf{x}) : \Re^{|A|} \to \Re^+$ and thus, the reliability can be defined as $R(\mathbf{x}) = P(\varphi(\mathbf{x}) \ge d)$ where d represents a network performance requirement.

For any network behavior, the reliability of a system for a given s–t requirement d in $G(N,A)$ can be calculated via: (1) analytic methods by enumerating all minimum paths/cuts or (2) applying one of the various simulation algorithms available. It should be noted that the exact calculation of all minimal cut sets is NP hard with computational effort growing exponentially with the number of nodes and arcs in the network. Thus, because of this limitation, in this paper, simulation has been used to estimate $R(\mathbf{x}) = P(\varphi(\mathbf{x}) \ge d)$.

Eq. (1) describes Category I resiliency as a probability density function of two-terminal network reliability when considering α external failures affecting the network and where the index β describes the specific α-failure case scenario considered within the network:

$$\mathcal{R}_{(\alpha,\beta)} = P(\varphi(\mathbf{x}) \ge d | \alpha, \beta) \forall \beta \tag{1}$$

Eq. (1) provides a profile of the behavior of the network's ability to satisfy a flow requirement d given α failures and thus, satisfies the concern that Category I resiliency should describe the sensitivity of network services to abnormal/external influences. Note that, the specific network resiliency $\mathcal{R}_{(\alpha,\beta)}$ is calculated for each of the unique non-repeating combination β to produce a distribution at a given α.

Once Eq. (1) is completely characterized for a given value of α relevant statistics can be obtained to gain insights regarding Category I network resiliency. For example, Eq. (2) provides the expected resiliency of the network in the presence of α failures:

$$\mathcal{R}_{\alpha} = \sum_{\beta} P(\varphi(\mathbf{x}) \ge d | \alpha, \beta) P(\beta) \tag{2}$$

3.1.1. Step-by-step illustration

To clearly describe the implementation of Eq. (1), consider the simple two-terminal network presented in Fig. 1. For this network, one can examine values for $\alpha = 1, 2, 3$. When $\alpha = 1$, there are three unique non-repeating combinations (i.e. $\beta = 1, 2, 3$) of a single arc failure. In state vector format, $((s,1),(1,t),(s,t))$, these combinations correspond to (0, 1, 1), (1, 0, 1), (1, 1, 0). This process is repeated for $\mathcal{R}_{(2,\beta)}$ and $\mathcal{R}_{(3,\beta)}$. At $\alpha = 3$, all the components are failed and thus, Category I resiliency of the network equals 0. The results for the first two cases are described in Table 1.

3.2. Simulation approach for resiliency computation

The results described in Table 1 are straightforward to obtain due to the size of the network. However, computing the exact

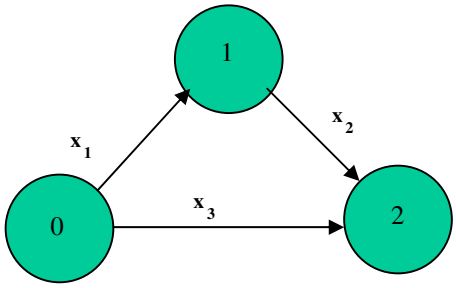


Fig. 1. Illustrative two-terminal network.

Table 1
Computations for $\mathcal{R}_{1(\beta)}$ and $\mathcal{R}_{2(\beta)}$.

β	$\mathbf{x}\|(1,\beta)$	$\mathcal{R}_{(1,\beta)}$	β	$\mathbf{x}\|(2,\beta)$	$\mathcal{R}_{(2,\beta)}$
1	(0, 1, 1)	0.7	1	(0, 0, 1)	0.7
2	(1, 0, 1)	0.7	2	(1, 0, 0)	0.0
3	(1, 1, 0)	0.81	3	(0, 1, 0)	0.0

Category I resiliency becomes highly intensive for large and complex networks due to growth of the unique non-repeating combinations, β. To address this issue, this paper proposes a Monte-Carlo simulation approach for approximating two-terminal network resiliency.

For a specific component failure scenario α, the approach presented in the simulation routine generates an estimate of resiliency by constructing a set of failure cases $(\alpha,\ \beta)$ and then, evaluating each case for their reliability. Initially, one calculates the maximum number of failure cases that will be considered, $\alpha_{\max}$. For each possible α, step 2 generates all possible components failure combination cases β and, in step 3, map such failure cases to a set of state vectors W through the structure mapping function $\Phi(Y, A)$. Then, reliability is obtained for each state vector $\mathbf{w}$ (where $\mathbf{w}$ is associated with a unique (α,β) case) using Monte-Carlo simulation.

Steps 4 through 11 produce the distribution for each given value of α. In step 5, a state vector is generated by using random integers to index into the possible state space variable values. Step 6 clears a counter variable q, which is used to count the random state vectors that satisfy the resiliency definition. Step 7 through 10 iterate through each state vector $\mathbf{x}$. Step 8 applies the failure case $\mathbf{w}$ (in vector form) to mask the components in $\mathbf{x}$, which produces $\mathbf{x}$ as a random state vector with failed components. In step 9, Eq. (1) is used to evaluate if $\mathbf{x}$ represents a resilient case. If so, step 10 increments counter variable q. Step 11 computes the expected value of the resiliency over the random states in X. Step 12 constructs a distribution of the expected values of resiliency for all the failure cases (α,β). Step 13 produces a set of distributions that then characterize the resiliency of the system for each α. The simulation routine pseudo-code follows:

Simulation Routine
Given:

$G(N,A)$ as network under test,
d is desired demand (for binary, $d = 1$)
$\alpha_{\max}$ is maximum α to test to, $\alpha_{\max} < |A|$
$M(m,n)$ is a uniform random generator,
$m \le M(m,n) \le n$

Compute $\mathcal{R}$:

1. $\forall \alpha: 1 \le \alpha \le \alpha_{\max}$,
2. $Y \leftarrow$ Comb $(|A|,\ \alpha)$

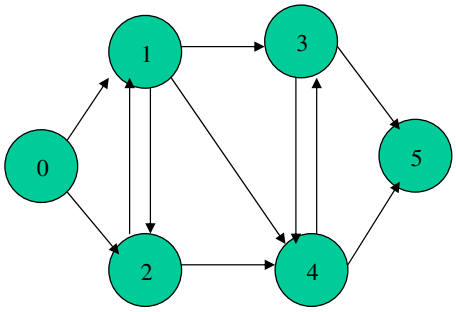


Fig. 2. The ARPA network.

Table 2
Reliability for arcs in ARPA network.

Arc	Reliability
0-1	0.90
0-2	0.90
1-2	0.90
2-1	0.90
1-3	0.60
1-4	0.60
2-4	0.99
3-4	0.90
4-3	0.90
3-5	0.90
4-5	0.90

3. $W \leftarrow \Phi(Y, A)$
4. $\forall w: w \in W$,
5. $X \leftarrow \{\mathbf{x}_i: x_{ij} = b_{jk}\}$, $k \in M(0, |\mathbf{b}_j|) \forall i,j$
6. $q = 0$
7. $\forall \mathbf{x}: \mathbf{x} \in X$,
8. $\underline{\mathbf{x}} = \mathbf{x} \cdot \mathbf{w}$
9. if $\varphi(\underline{\mathbf{x}}) \geq (d)$ then
10. $q = q+1$
11. $Я_w = q/|X|$
12. $Я_z = \{Я_w\}$, $w \in W$
13. $Я = \{Я_z\}, \alpha: 1 \leq \alpha \leq \alpha_{\max}$
    End.

4. Network analysis

To test the implementation of the equations described in Section 3, four networks were analyzed. The first three are binary networks with varying topology while the last network is multi-state.

4.1. ARPA network

The "ARPA" network is shown in Fig. 2. This network has been previously analyzed in literature from a reliability perspective [5]. For this two-terminal binary network, it is assumed that node 0 is the source, and node 5 is the sink. The reliability of each arc is given in Table 2.

For this network, Fig. 3 illustrates Category I resiliency of the network for each possible value of α. For each of the different

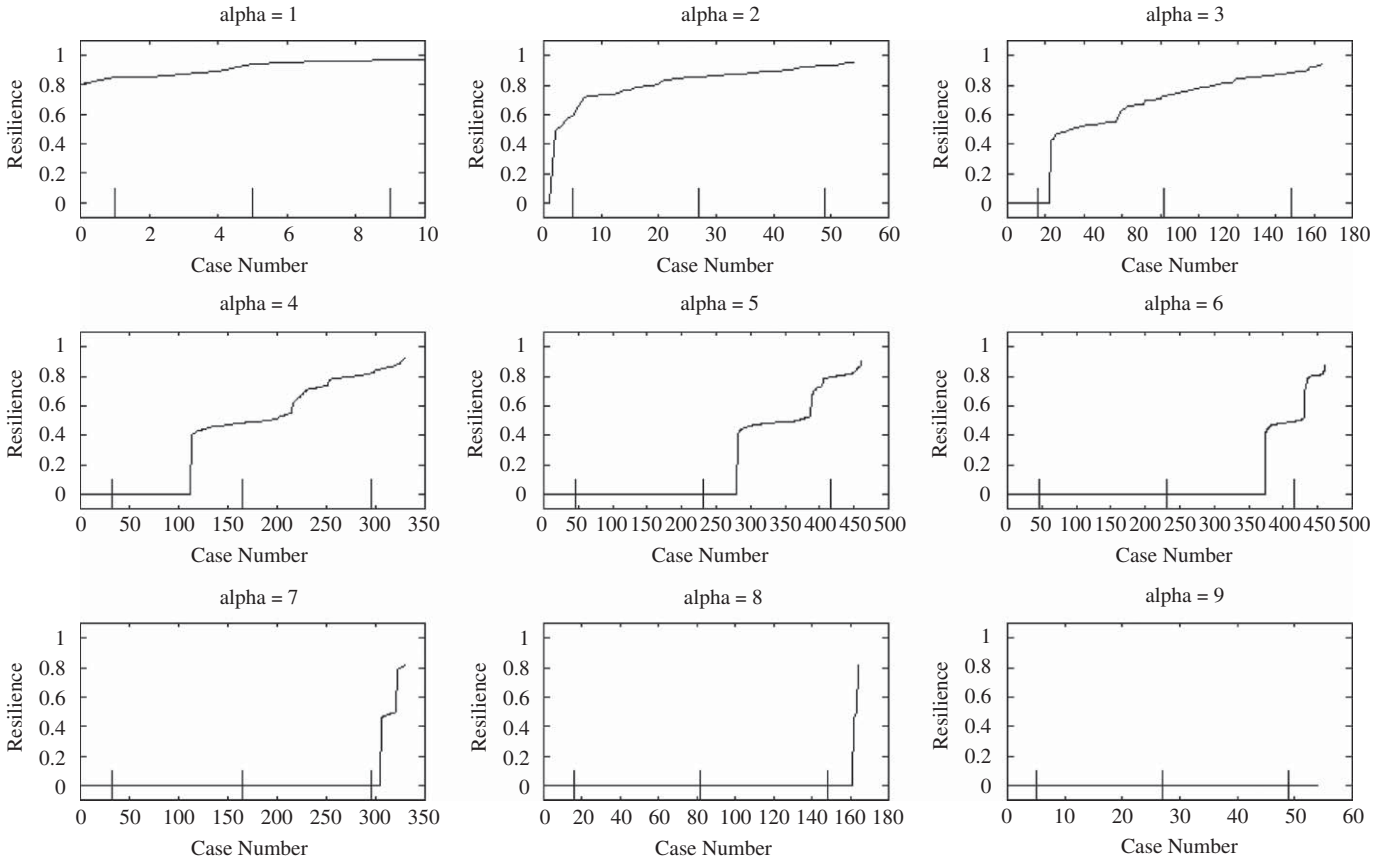


Fig. 3. Resiliency behavior for all α cases.

graphs presented in this figure, the marked "ticks" represent 10%, 50%, 90% population thresholds. Moreover, Table 3 presents insightful statistics about the behavior of the network.

Based on Fig. 3 and Table 3 the following insight can be obtained:

When $\alpha = 1$, Category I resiliency is equivalent to Fussell–Vesely (FV) [32] component importance measure. As described, it shows that no single failure can trigger a catastrophically network failure. With respect to the threshold "ticks', one can infer that: (1) only 10% of the failures cases have a reliability below 0.8020, (2) conversely, only 10% of the cases have a reliability higher than 90% and finally, (3) 50% of the failure cases have a reliability higher than 0.9430. Thus, the ARPA network can be said to have high Category I resiliency to this value of α.

Furthermore, when $\alpha = 1$, Category I resiliency can be used to rank the components in terms of their damage to network reliability. As such, Table 4 illustrates the comparison of $\mathcal{R}_{(1,\beta)}$ against FV [32] and Birnbaum [8] CIM. Based on this table, the rankings of the components illustrates that $\mathcal{R}_{(1,\beta)}$ corresponds with FV.

Before discussing the remaining values for α, it is important to clarify that the purpose Category I resiliency is to describe the sensitivity of the network performance to abnormal/external influences. Such purpose establishes $\mathcal{R}_{(\alpha,\beta)}$ as a CIM. However, for values of α greater than 1, the ranking of network components based on $\mathcal{R}_{(\alpha,\beta)}$ is not trivial and requires further research that although outside the scope of this manuscript the authors are pursuing. Nevertheless, for such values of α, Category I resiliency can identify the cases β that have the highest impact on network performance.

Out of the 55 possible failure cases for $\alpha = 2$, there are two distinct cases where the network can fail: if the source node has no exiting arcs (i.e. both arcs 0-1 and 0-2 are failed) or the sink node has no entering arcs (i.e. both arcs 3-5 and 4-5 are failed). Note however that more than 50% of the cases guarantee that network reliability will be greater than 0.8530.

Table 3
Statistics for ARPA network resiliency.

α	Minimum	Mean	Median	Maximum	10% Threshold	90% Threshold
1	0.8140	0.9121	0.9430	0.9733	0.8020	0.9650
2	0.0000	0.7955	0.8530	0.9570	0.7260	0.9320
3	0.0000	0.6252	0.7260	0.9390	0.5260	0.8920
4	0.0000	0.4210	0.4810	0.9190	0.4610	0.8610
5	0.0000	0.2380	0.0000	0.9010	0.4670	0.8220
6	0.0000	0.1128	0.0000	0.8720	0.4670	0.8120
7	0.0000	0.0427	0.0000	0.8100	0.4660	0.8010
8	0.0000	0.0107	0.0000	0.8150	0.0000	0.4920
9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 4
Comparison of $\mathcal{R}_{(1,\beta)}$ with FV [32] and Birnbaum [8].

β	Arc	$\mathcal{R}_{(1,\beta)}$	FV	Birn
1	2-4	0.814	0.164	0.162
2	4-5	0.853	0.123	0.134
3	0-2	0.875	0.101	0.112
4	0-1	0.882	0.094	0.099
5	3-5	0.888	0.088	0.093
6	4-3	0.944	0.031	0.034
7	1-2	0.960	0.014	0.015
8	1-3	0.964	0.009	0.011
9	1-4	0.970	0.004	0.005
10	2-1	0.972	0.001	0.001
11	3-4	0.973	0.000	0.000

For $\alpha = 3$ the network experiences more catastrophic failures. Out of the total 165 unique failure cases, 22 correspond to a total network failure. This information from Category I resiliency is extremely important for a future characterization of Category II resiliency. Note that the median Category I resiliency is reduced by roughly 15% when compared to the $\alpha = 2$ case.

The system degrades as more failures are introduced. For $\alpha = 4$, 112 of the cases represent a total network failure from the 330 unique failure cases. The sharp rise from zero to 0.4030 in resiliency reflects the small number of components in this network. However, two thirds of the time the network will provide an adequate level of service. For $\alpha = 5$, of 462 cases now, 281 represent catastrophic failure, illustrating that the network is completely failed for more than half the failure cases.

For $\alpha = 6$, a total of 374 of 462 cases are now complete failures. Note that for $\alpha = 6$, now only five components can be considered to be working and, as a result, the count of the potential failure cases begins to expand. For $\alpha = 7$, 307 of 330 cases are total network failures while for $\alpha = 8$, 161 of 164 cases are failed. In this latter case, the three non-failed cases that allow the network to function correspond to minimal path sets of the network. Finally, for $\alpha = 9$, there are no cases that allow the network to function. This makes intuitive sense, as there are no visible solutions that only use two arcs to provide a path through the network.

4.2. Feed-forward network

The second network analyzed has been randomly generated as a feed-forward network with $|A| = 31$. This network was generated by randomly placing nodes on a grid and connecting nodes in a forward (left to right) direction Similarly, arc reliabilities were generated randomly as illustrated in Table 5. Due to the computational complexity in this network, α was evaluated up to a value of $\alpha_{\max} = 5$. The illustration of the network is presented in Fig. 4.

As described by both Fig. 5 and Table 6, and similar to the ARPA network, the feed-forward network starts exhibiting catastrophic failures when the source and sink nodes are cutoff from the rest of the network (i.e. cases that correspond to network cut sets).

For $\alpha = 1$, out of 30 failure cases this network contains no catastrophic failures. This information alone indicates that there is sufficient redundancy in the network to accommodate a single component failure and still guarantee system functioning (i.e. $\varphi(x) = 1$). For $\alpha = 2$, out of 465 failure cases analyzed, there is exactly one case that resulted in complete network failure (roughly 0.2% of the total population) Also, it is important to note

Table 5
Arc reliability for feed-forward network.

Arc	Reliability	Arc	Reliability
1	0.9308	16	0.8312
2	0.8194	17	0.8214
3	0.9968	18	0.9193
4	0.7221	19	0.9214
5	0.8511	20	0.897
6	0.9053	21	0.8774
7	0.7689	22	0.9665
8	0.9079	23	0.9342
9	0.6538	24	0.7862
10	0.9341	25	0.9937
11	0.8749	26	0.8967
12	0.8997	27	0.5723
13	0.9335	28	0.9596
14	0.9785	29	0.9659
15	0.9767	30	0.9923
		31	0.765

that more than 90% of the cases exhibit a reliability higher than 80%. For $\alpha = 3$, of 4495 failure cases there are 34 (0.7%) that resulted in a total failure. However, as α increases the resiliency of the network decreases as evidenced by the fact that roughly 15% of the failure cases correspond to a reliability of less than 80%. When $\alpha = 4$, of 31,464 failure cases, 556 (1.7%) resulted in complete network failure. Finally, for $\alpha = 5$ there were 169,911 failure cases of which 5815 (3.4%) resulted in catastrophic failure.

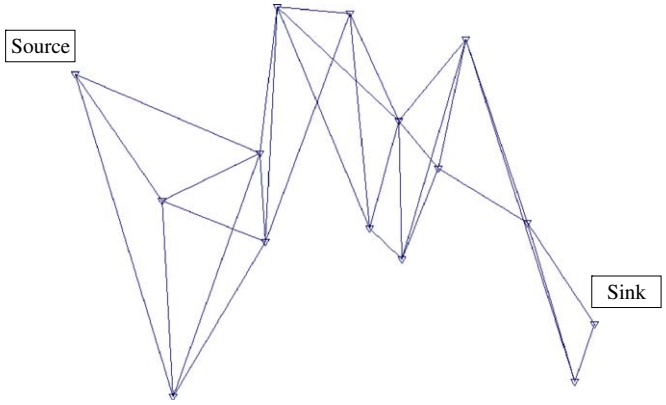


Fig. 4. Feed-forward network with 31 arcs.

4.3. Mesh network—binary and multi-state case

4.3.1. Binary case

The third network considered in this research manuscript is a spatially distributed mesh network configured as $|N| = 20$ and $|A| = 100$, with each node connected outward to its five nearest neighbors. The leftmost and rightmost nodes in the graph determined the source and the sink nodes. As illustrated in Fig. 6, this network has high redundancy and is imitative of a MANET or other distributed communications networks. Moreover,

Table 6
Statistics for feed-forward network resiliency.

α	Minimum	Mean	Median	Maximum	10% Threshold	90% Threshold
1	0.7590	0.9670	0.9810	0.9900	0.9370	0.9860
2	0.0000	0.9451	0.9720	0.9930	0.8840	0.9840
3	0.0000	0.9158	0.9570	0.9940	0.8390	0.9810
4	0.0000	0.8787	0.9340	0.9930	0.7560	0.9750
5	0.0000	0.8326	0.8950	0.9950	0.7290	0.9680

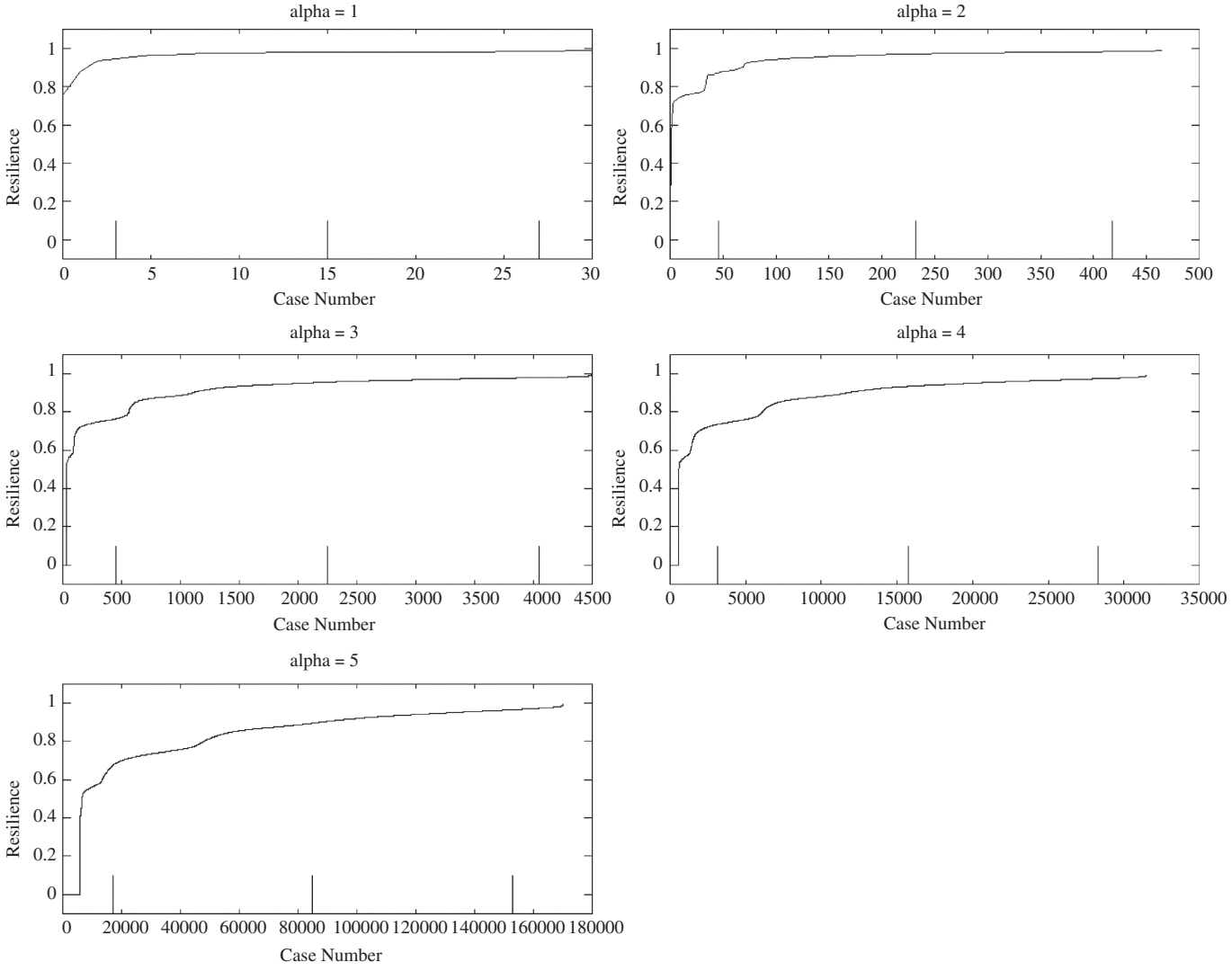


Fig. 5. Feed-forward network resiliency behavior for $\alpha \leq \alpha_{max}$ cases.

this network is homogenous in the sense that all the arcs have identical reliability equal to 0.90. Finally, due to the size of this network $\alpha_{max} = 3$.

As illustrated by Table 7, the network is very resilient for $\alpha = 1$, 2, 3 due to its redundancy. However, it is interesting to note that one can identify a small variation in resiliency for $\alpha = 3$, indicating that some combination of three failed arcs and the normal reliabilities of the operating arcs created a set of conditions that caused a significant drop in minimum reliability—from 0.9810 to 0.8500. A detailed analysis showed that the worst case consisted of three failed arcs that connected to one central node that was not the source or the sink node.

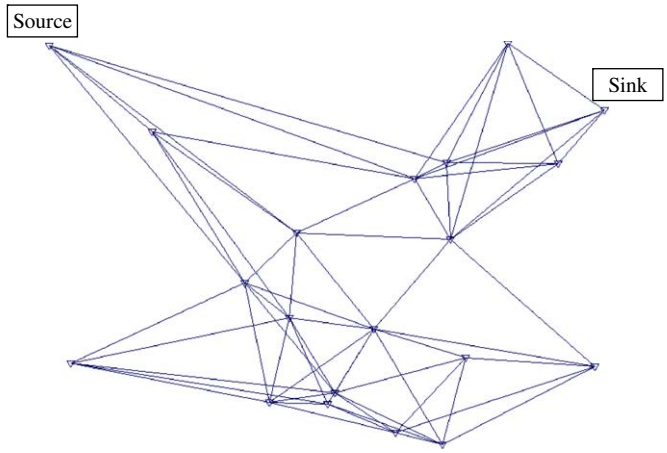


Fig. 6. Homogeneous mesh network with 100 arcs.

Table 7
Statistics for binary mesh network resiliency.

α	Minimum	Mean	Median	Maximum	10% Threshold	90% Threshold
1	0.9970	0.9998	1.0000	1.0000	0.9990	1.0000
2	0.9810	0.9996	1.0000	1.0000	0.9990	1.0000
3	0.8500	0.9994	1.0000	1.0000	1.0000	1.0000

4.3.2. Multi-state case

The final analysis for this section considers the network described in Fig. 6 but assuming it follows a multi-state network behavior. Each arch in the network is assumed to have the following capacity and associated state probability: (0, 0.05), (1, 0.10) and (2, 0.85).

For this new set up, the concept of resiliency offers considerable insight to the behavior of the mesh network. Since this is a multi-state network, $\mathcal{R}_{(\alpha,\beta)}$ is evaluated using the multi-state form, $\mathcal{R}_{(\alpha,\beta)} = P(\varphi(\mathbf{x}) \geq d | (\alpha,\beta))$. The requirement d was set to a value of 6 units. Whenever the network satisfies this requirement, then it considered resilient for that specific failure case α. Note that the maximum capacity of any individual arc is 2, so that meeting a demand $d = 6$ must involve load sharing among at least three arcs.

The change between Category I resiliency for the multi-state and binary cases is remarkable as illustrated by Fig. 7 and Table 8. For $\alpha = 1$, the binary case showed no major perturbation to the functioning of the system. However, when considering a multi-state perspective, roughly 10% of the cases report reliability below 60%. Moreover, it can be observed that when $\alpha = 2$ the network experiences catastrophic failures. These scenarios correspond to 23 cases out of a possible 4949. Finally, for $\alpha = 3$ there are 2306 out of 161,700 cases that cause the network to be unable to meet the requirement $d = 6$.

The multi-state case illustrates the difference between networks that require load sharing (i.e. multi-state systems) and binary systems. The mesh network in the binary case did not support load sharing and thus, if there was at least one set of components that provided a path from source to sink, then the network was considered connected. In other words, in the binary case, there is a great amount of redundancy that allows the network to be regarded as highly Category I resilient to 1, 2, and 3

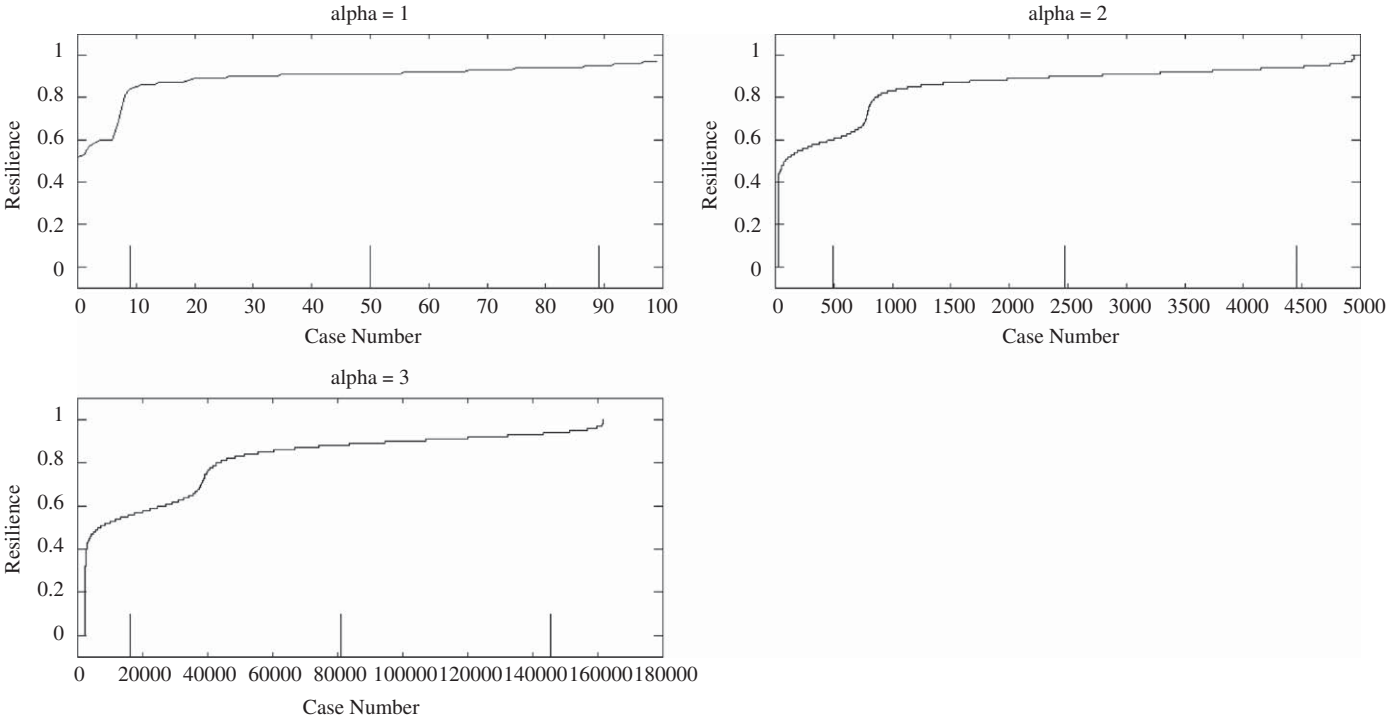


Fig. 7. Resiliency for $\alpha \leq \alpha_{max} = 3$ and $d = 6$.

Table 8
Statistics for multi-state mesh network resiliency.

α	Minimum	Mean	Median	Maximum	10% Threshold	90% Threshold
1	0.5200	0.8886	0.9100	0.9700	0.8600	0.9500
2	0.0000	0.8484	0.9000	1.0000	0.6700	0.9400
3	0.0000	0.8083	0.8800	1.0000	0.6000	0.9400

external failures. However, in the multi-state mesh network, aggregate capacity became mandatory based on the requirement that exceeds any individual component capacity. When aggregate components are required to function simultaneously, the network sensitivity to multiple simultaneous component failures increases (i.e. Category I resiliency decreases).

5. Conclusion

This paper formalized a definition and procedure for calculating Category I resiliency. The experimental results on networks with different topology, complexity and behavior illustrate that the proposed equations can be applied as a component importance measure to provide a better understanding of the performance, strengths and weaknesses of networks for binary and multi-state capacitated networks when in the presence of as external causes of component failure. Such knowledge is of great concern to networks that are vulnerable to abnormal failure conditions and varying load conditions, such as those in communication and distribution networks with spatial dependencies.

The interpretation of the system resiliency distributions is clear for cases where the number of as external causes of component failure is low. In particular, when $\alpha = 1$ Category resiliency is equivalent to FV. For higher values of α, the authors of this paper recognize that providing an exact calculation for resiliency is an NP-hard problem. Thus, future research will be devoted into statistically analyzing the appropriate number of (α,β) cases that need to be analyzed to accurately approximate $\mathcal{R}_{(\alpha,\beta)}$. That is, for larger values of α, approaches to identify specific combinations of component failure that are statistically significant in their impact on the network performance. There is also a need for future methods to use Category I resiliency as a metric to compare network alternate configurations. Useful comparisons could include design optimization as well as tradeoff analysis between alternate solutions to specific design (re)configuration problems.

References

[1] Serway R. Mechanics of solids and fluids. In: Physics for scientists and engineers with modern physics. second edition. New York, NY: Saunders College Publishing; 1986.

[2] Westmark V. A Definition for Information System Survivability. In: IEEE Proceedings of the 37th Hawaii International Conference on System Sciences, 2004.

[3] Rose A. Economic resilience to natural and man-made disasters: Multidisciplinary origins and contextual dimensions. Environmental Hazards 2007;7(4):383–98.

[4] Gebre B, Ramirez-Marquez JE. Element substitution algorithm for general two-terminal network reliability analyses. IIE Transactions 2007;39:265–75.

[5] Ramirez-Marquez JE, Coit D. A Monte-Carlo simulation approach for approximating multi-state two-terminal reliability. Reliability Engineering and System Safety 2005;87(2):253–64.

[6] Ramirez-Marquez JE, Assefa B. A classification tree based approach for the development of minimal cut and path vectors of a network. IEEE Transactions on Reliability 2007;56(3):474–87.

[7] Assefa B, Ramirez-Marquez JE. Element substitution algorithm for general two-terminal network reliability analyses. IIE Transactions 2007;39(3):265–75.

[8] Elsayed E. Reliability engineering. Reading, MA: Addison Wesley Longman Inc; 1996.

[9] Rausand M, Hoyland A. Component importance. In: System reliability theory. New York, NY: Wiley; 2004.

[10] Ramirez-Marquez JE, Coit D. Multistate component criticality analysis and reliability prioritization in multistate systems. Reliability Engineering and System Safety, Special Issue on ESREL 2005 2007;92(12):1608–19.

[11] Ramirez-Marquez JE, Coit DW. Composite importance measures for multi-state systems with multi-state components. IEEE Transactions on Reliability 2005;54(3):2005.

[12] Zio E, Podofillini L. Importance measures and genetic algorithms for designing a risk-informed optimally balanced system. Reliability Engineering and System Safety 2007;92(10):1435–47.

[13] Zio E, Marella M, Podofillini L. Importance measures-based prioritization for improving the performance of multi-state systems: application to the railway industry. Reliability Engineering and System Safety 2007;92(10):1303–14.

[14] Levitin G, Lisnianski A. Importance and sensitivity analysis of multi-state systems using the universal generating function method. Reliability Engineering and System Safety 1999;65(3):271–82.

[15] Lee J, Chapin SJ, Taylor S. Reliable heterogeneous applications. IEEE Transactions on Reliability 2003;52(3).

[16] Song J, Der Kiureghian A. Component importance measures by linear programming bounds on system reliability. In: The 9th international conference on structural safety and reliability (ICOSSAR9), Rome, Italy, June 19–23, 2005.

[17] Vaurio JK. Extensions of the uncertainty quantification of common cause failure rates. Reliability Engineering and System Safety 2002;78:63–9.

[18] Vaurio JK. The probabilistic modeling of external common cause failure shocks in redundant systems. Reliability Engineering and System Safety 1995;50:97–107.

[19] Butler DA. An importance ranking for components based upon cuts. Operations Research 1977;25:874–9.

[20] Podofillini L, Zio E. Events group risk importance by genetic algorithms. Journal of Risk and Reliability 2008;222(3).

[21] Erik Hollnagel, David D. Woods, Nancy Leveson. Resilience engineering: concepts and precepts. Ashgate Publishing, Ltd., 2006, ISBN: 0754646416.

[22] Castet J, Saleh J. Survivability and resiliency of spacecraft and space-based networks: a framework for characterization and analysis. American Institute of Aeronautics and Astronautics, AIAA Technical Report 2008-7707.

[23] She Q, Huang X, Jue JP. Maximum survivability under multiple failures. 2006 National Fiber Optic Engineers Conference, 5–10 March 2006.

[24] Banner R, Orda A. The power of tuning: a novel approach for the efficient design of survivable networks. IEEE/ACM Transactions on Networking 2007;15(4):737–49.

[25] Pullan W. Optimising multiple aspects of network survivability. In: Proceedings of the 2002 congress on evolutionary computation, vol. 1, 12–17 May 2002. p. 115–20.

[26] Zolfaghari A, Kaudel FJ. Framework for network survivability performance. IEEE Journal on Selected Areas in Communications 1994;12(1):46–51.

[27] Bu T, Chan MC, Ramjee R. Connectivity, performance, and resiliency of IP-based CDMA radio access networks. IEEE Transactions on Mobile Computing 2006;5(8):1103–18.

[28] Liew SC, Lu KW. A framework for characterizing disaster-based network survivability. IEEE Journal on Selected Areas in Communications 1994;12(1):52–8.

[29] Kalyoncu H, Sankur B. Estimation of survivability of communication networks. Electronics Letters 1992;28(19):1790–1.

[30] Gautam BR, Schulz NN, Srivastava AK. Developing a survivability index for distribution systems including islanding. In: 39th North American power symposium, September 30 2007–October 2, 2007. p. 595–9.

[31] Al-Ammar E, Fisher J. Resiliency assessment of the power system network to cyber and physical attacks. IEEE power engineering society general meeting, 18–22 June 2006.

[32] Fussell J. How to calculate system reliability and safety characteristics. IEEE Transactions on Reliability 1975;24(3):169–74.